

# 2023-10-13 - Knowledge Representation and Reasoning - Data Quality

Cursus: B-CS Knowledge Representation and Reasoning & Data Quality - 202200022 – 2023-2024

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This exam is closed book, you may only use a non-graphical calculator (available in Remindo), and a 1 A4 cheat sheet.

This examination consists of multiple-choice and open questions; the DQ part of this exam should take 1 hour.

Tips:

- Read each question carefully without looking at the possible answers.
- If you can answer in a freetext box, do volunteer an explanation, because also partially correct answers may be worth some points.
- Do not stay on any one question for too long. If you do not know the answer and have spent more than 10 minutes on the question, move on to the next question and come back to this one later.
- If you have any time left over at the end, check your answers.

If you have questions about the content of the exam, you can contact a TA or teacher in the room.

In total you can obtain a maximum of 90 points for this test. You need at least 45 points to pass the test (grade  $\geq 5.5$ ).

Good luck!

**Aantal vragen: 3**

**In totaal zijn 90 punten voor deze toets te behalen, 48,06 punten zijn nodig om voor de toets te slagen.**

## Casus 1

This example database is inspired by the strategy game "Forge of Empires". It contains information about Great Buildings that you can build in this game. Some active player in this gaming community built a web-based tool to help other players with choosing which Great Buildings to build or upgrade to a higher level. The database below is used by this tool and contains relevant facts about the Great Buildings to be used by the tool.

### GreatBuildings

| bid | name           | age         | level | cost | bonus | size | descr                                           |
|-----|----------------|-------------|-------|------|-------|------|-------------------------------------------------|
| 1   | Tower of Babel | Bronze      | 1     | 40   | 6     | 4x4  | Increases population and produces goods         |
| 2   | Tower of Babel | Bronze      | 2     | 60   | 7     | 4x4  | Increases population and produces goods         |
| 3   | Collosseum     | Iron        | 1     | 40   | 10    | 6x7  | Increases happiness and produces medals         |
| 4   | Collosseum     | Iron        | 2     | 60   | 11    | 7x8  | Increases happiness and produces medals         |
| 5   | Alcatraz       | Progressive | 1     | 60   | 4     | 10x7 | Increases happiness and produces military units |
| 6   | Alcatraz       | Progressive | 2     | 90   | 4     | 10x7 | Increases happiness and produces military units |
| 7   | Alcatraz       | Progressive | 3     | 170  | 5     | 10x7 | Increases happiness and produces military units |

### Description of the attributes

|       |                                                                                                                                |
|-------|--------------------------------------------------------------------------------------------------------------------------------|
| bid   | Building ID. Automatically generated.                                                                                          |
| name  | The name of the Great Building.                                                                                                |
| age   | The age the Great Building belongs to. One can only build the Great Building after having progressed to this age in the game.  |
| level | Great Buildings can be upgraded to higher levels. This comes at a cost but produces more bonus (see the two attributes below). |
| cost  | The cost to upgrade the Great Building to this level (provided that it was on one level below it).                             |
| bonus | The bonus that is received every day for the Great Building at this level.                                                     |
| size  | The size of the Great Building in terms of grid dimensions. The notation NxM means N by M grid tiles.                          |
| descr | Description of the Great Building.                                                                                             |

## Constraints

- 7 pt. a. **Question Functional dependencies 1 <Easy>**  
Should the functional dependency "bid  $\rightarrow$  name,age,size" hold for this database? Explain your answer.  
**NB:** Do not look at the data; only base your answer on the description of the attributes and your knowledge of how social media works
- 7 pt. b. **Question Functional dependencies 2 <Easy>**  
Should the functional dependency "bid,level  $\rightarrow$  cost,bonus,size" hold for this database? Explain your answer.  
**NB:** Do not look at the data; only base your answer on the description of the attributes and your knowledge of how social media works
- 3 pt. c. **Question Formal notation <Reg>**  
Which is the correct formal notation for the constraint "bid,level  $\rightarrow$  cost"
- This question was cancelled because none of the answers is perfectly correct
- a.  $\forall a1 \forall b1 \forall c1 \forall d1 \forall e \forall f1 \forall g1 \forall h1 \forall a2 \forall b2 \forall c2 \forall d2 \forall f2 \forall g2 \forall h2 ( \text{GreatBuildings}(a1,b1,c1,d1,e,f1,g1,h1) \wedge \text{GreatBuildings}(a2,b2,c2,d2,e,f2,g2,h2) \Rightarrow (a1=a2 \wedge b1=b2) )$
- b.  $\forall a1 \forall a2 \forall b1 \forall b2 \forall c \forall d \forall e \forall f \forall g \forall h \forall i ( \text{GreatBuildings}(a1,b1,c,d,e,f,g,h) \wedge \text{GreatBuildings}(a2,b2,c,d,e,f,g,h) \Rightarrow (a1=a2 \wedge b1=b2) )$
- c.  $\forall a \forall b \forall c \forall d \forall e1 \forall e2 \forall f \forall g \forall h ( \text{GreatBuildings}(a,b,c,d,e1,f,g,h) \wedge \text{GreatBuildings}(a,b,c,d,e2,f,g,h) \Rightarrow e1=e2 )$
- d.  $\forall a \forall b \forall c1 \forall d1 \forall e1 \forall f1 \forall g1 \forall h1 \forall c2 \forall d2 \forall e2 \forall f2 \forall g2 \forall h1 ( \text{GreatBuildings}(a,b,c1,d1,e1,f1,g1,h1) \wedge \text{GreatBuildings}(a,b,c2,d2,e2,f2,g2,h2) \Rightarrow e1=e2 )$
- 7 pt. d. **Question Functional dependencies 3 <Easy>**  
Does the functional dependency "bid  $\rightarrow$  age,size" hold for the system? Explain your answer.  
**NB:** Now look at the data.

- 6 pt. e. **Question Data Quality <Easy>**  
Imagine an information system in a company which stores its data in a database. Which of the following statements about "Data Quality" for the data in this database are true? (multiple answers possible)
- a. Given that the data in this information system's database is of high quality, it can be effectively used for business analytics.
  - b. There are a lot of missing values in this database, but it always contains sufficient information for the users of the information system for their tasks, so despite this issue, the quality of the data is high.
  - c. The documentation of the database is bad. This can be a reason for bad data quality.
  - d. A new version of the information system software is released because of changed requirements. As a consequence, the data in the database was all of a sudden of higher quality.
- 3 pt. f. **Question Repairs <Reg>**  
Given one table and one constraint for which there is one violation in the table, what is true for the number of repairs?
- a. The number of repairs can be infinite
  - b. There is either no repair or exactly one.
  - c. There is exactly one repair.
  - d. The number of repairs is always finite.
- 4 pt. g. **Question Consistent Query Answering (CQA) <Reg>**  
What is the result of the query below under the Consistent Query Answering approach given that the constraint "bid  $\rightarrow$  name,age,size" should hold? We consider insertions and deletions as possible repairs. Explain your answer.
- ```
SELECT DISTINCT cost
FROM GreatBuildings
```

5 pt. h. **Question Answer Set Programming 1 <Adv>**

For brevity reasons, we use the name "GB" for the "GreatBuildings" table. Moreover, we only consider now the attributes ('bid', 'name', 'age', 'level', and 'size') and only the constraint  $\text{bid} \rightarrow \text{name, age, size}$ . Below is an answer set programme given to find possible repairs for violations of this constraint. We use the variables B, N, A, L, and S for the attributes 'bid', 'name', 'age', 'level', and 'size', respectively.

Consider the following answer set program:

1.  $\text{GB\_}(B1, N, A, L1, S, f) \vee \text{GB\_}(B2, N, A, L2, S, f) \leftarrow \text{GB\_}(B1, N, A, L1, S, t^*) \wedge \text{GB\_}(B2, N, A, L2, S, t^*) \wedge B1 \neq B2$
2.  $\text{GB\_}(B, N, A, L, S, t^*) \leftarrow \text{GB}(B, N, A, L, S)$
3.  $\text{GB\_}(B, N, A, L, S, t^*) \leftarrow \text{GB\_}(B, N, A, L, S, t)$
4.  $\text{GB\_}(B, N, A, L, S, t^{**}) \leftarrow \text{GB\_}(B, N, A, L, S, t^*) \wedge \neg \text{GB\_}(B, N, A, L, S, f)$
5.  $\leftarrow \text{GB\_}(B, N, A, L, S, t) \wedge \text{GB\_}(B, N, A, L, S, f)$

For your convenience, here are the descriptions of the annotations used in the answer set program above:

**t**: repair by insertion of the tuple in the new table

**f**: repair by deletion of the tuple from the new table

**t\***: tuple was in the original table or inserted in the repair process

**t\*\***: tuple belongs to final contents of the new table

Are all rules of this answer set program used when the program is applied to the given database? Explain your answer.

4 pt. i. **Question Answer Set Programming 2 <Adv>**

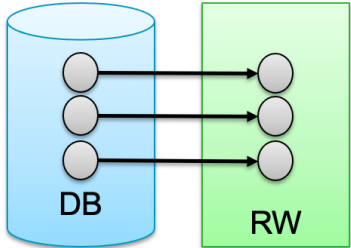
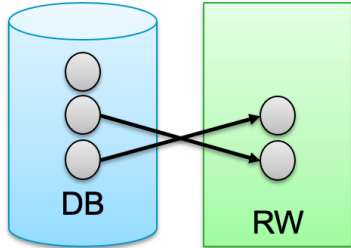
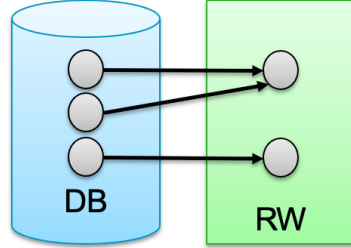
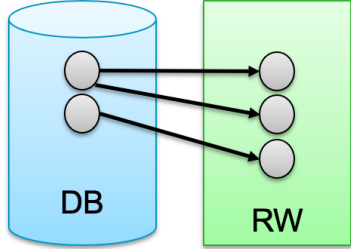
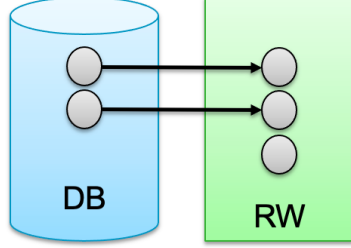
The program of the previous question generates repairs for the violation of the constraint, i.e., same bid but different values for name, age, or size). What possible minimal repairs does the program of the previous question include? (multiple answers possible)

- a. The possible minimal repair included in the program is to give one of the two records a different bid if a pair violates the constraint.
- b. The possible minimal repair included in the program is deleting one of the two records if a pair violates the constraint.
- c. The possible minimal repair included in the program is to fill in the bid, name, age, and size attributes if their values are missing.
- d. The possible minimal repair included in the program is changing the value of the size attribute in one of the two records if a pair violates the constraint, because size contains values that violates the constraint.
- e. The possible minimal repair included in the program is deleting both records if a pair violates the constraint.

## 2 Duplicates

7 pt. a. **Question** *Duplicate detection* <Easy>

Match each depiction of a particular patterns in the relationship between data in the database and the real world with the right term.

|                                                                                          | Proper<br>representation | Ambiguous<br>representation | Incomplete<br>representation | Duplicate             |
|------------------------------------------------------------------------------------------|--------------------------|-----------------------------|------------------------------|-----------------------|
|                                                                                          | A                        | B                           | C                            | D                     |
| <br>1   | <input type="radio"/>    | <input type="radio"/>       | <input type="radio"/>        | <input type="radio"/> |
| <br>2  | <input type="radio"/>    | <input type="radio"/>       | <input type="radio"/>        | <input type="radio"/> |
| <br>3 | <input type="radio"/>    | <input type="radio"/>       | <input type="radio"/>        | <input type="radio"/> |
| <br>4 | <input type="radio"/>    | <input type="radio"/>       | <input type="radio"/>        | <input type="radio"/> |
| <br>5 | <input type="radio"/>    | <input type="radio"/>       | <input type="radio"/>        | <input type="radio"/> |

**Question** *String similarity* <Adv>

The formula for Jaccard distance is  $\text{Jaccard}(P, Q) = \frac{|P \cap Q|}{|P \cup Q|}$ . We define 2-grams as tokens (i.e.,

a token is a string of 2 subsequent characters). We are interested in the similarity according to this distance metric for strings "Flower" and "Flwer". Below you find the skeleton of the full calculation of the Jaccard distance between these strings. Please complete it by first filling in what exactly P is for string "Flower" in this example; and then filling in the two numbers of the division and finally the end result.

P = **b.** ..... (4 pt.)

Q = [not necessary to give it]

$|P \cap Q|$  = **c.** ..... (1 pt.)

$|P \cup Q|$  = **d.** ..... (1 pt.)

Jaccard(P,Q) = **e.** ..... (2 pt.)

#### Question Clustering <Reg>

Suppose we did a duplicate matching step for a table of 5 records A through E. The matching scores can be found in the table below.

|   | A | B   | C    | D    | E    |
|---|---|-----|------|------|------|
| A | 1 | 0.1 | 0.15 | 0.9  | 0.7  |
| B |   | 1   | 0.6  | 0.2  | 0.1  |
| C |   |     | 1    | 0.05 | 0.1  |
| D |   |     |      | 1    | 0.75 |
| E |   |     |      |      | 1    |

We use thresholds  $T_L=0.4$  (lower threshold) and  $T_U=0.8$  (upper threshold).

4 pt. **f.** Give all possible partitionings for this duplicate matching result. Explain your answer.

4 pt. **g.** **Question Clustering <Reg>**

For one of the possible partitionings you came up with, calculate the probability of that partitioning.

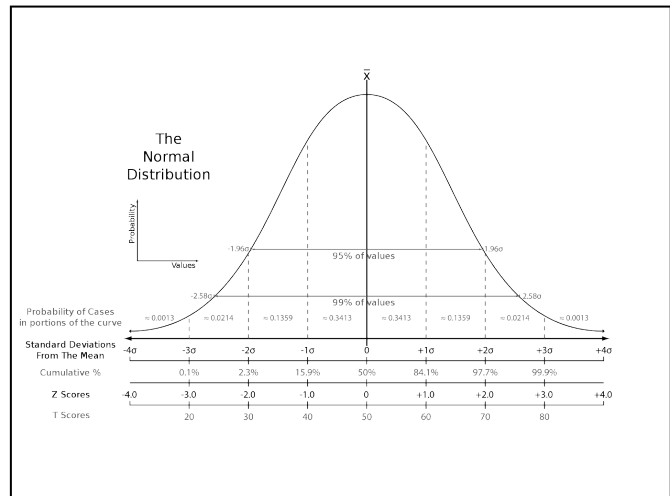
Explain your answer with a full calculation in multiple steps. If you don't have a calculator, just leave out the computation to a final number (i.e., a final answer like  $(0.4+0.25*0.89)/5$  is perfectly fine).

### 3 Other: Anomaly detection & Business Perspective

2 pt.

- a. **Question Z-score <Reg>**  
On the right you see the picture explaining the Z-score. Which of the following statements is true for the Z-score? (multiple answers possible)

- a. 0.2% of the values of an attribute have a Z-score lower than -3.0 or higher than 3.0



- b. An attribute value has a Z-score of 2.0 if it is 2.0 standard deviations larger than the mean
- c. A missing value is also an outlier
- d. Outliers have a Z-score near 0 (zero)
- e. An attribute value has a Z-score of 2.0 if it is 2.0 standard deviations larger than the average
- f. A Z-score doesn't make sense for nominal values

3 pt.

- b. **Question "AI in Health" motivational story of introductory lecture <Reg>**

In the motivational story about the struggle for high quality data for breast cancer diagnosis, it was explained that most predictive models developed for breast cancer had an input of 4 mammography images for predicting either malignant or benign (i.e., cancer vs. normal or harmless abnormality). The 4 images are: a top and side view for each breast.

What was the reason given why these models are not applicable in practice?

- a. In an IT infrastructure at a hospital, it is not possible to give at the same time 4 mammography images to an AI component
- b. A certain percentage (around 8%) of patients do not have exactly 4 mammography images and a hospital would not want to exclude these patients
- c. A model that has as input only one mammography image gives a better prediction performance
- d. Clinicians do not make a diagnosis based on 4 mammography images at the same time.
- e. The 4 mammography images cannot be aligned sufficiently in practice for such models to work properly, hence the quality of these mammography images is insufficient for this purpose

- 8 pt.    **c.**    **Question *Business Perspective* <Easy>**  
In the book chapters and on the slides, the claim was made that "there is little what tech departments can do to improve data quality". Explain (a) why can they do only so little, i.e., what can they not do, and (b) what they *can* do.
- 6 pt.    **d.**    **Question *Business Perspective* <Easy>**  
Give two examples of "cost of poor data quality" that are not about money.

## Correctiemodel

1.

47 pt.

a.

| Correction criterion         | Points          |
|------------------------------|-----------------|
| Notion of uniquely determine | 4 points        |
| Correct                      | 3 points        |
| <i>Total points:</i>         | <i>7 points</i> |

b.

| Correction criterion         | Points          |
|------------------------------|-----------------|
| Notion of uniquely determine | 4 points        |
| Correct                      | 3 points        |
| <i>Total points:</i>         | <i>7 points</i> |

c. 3 pt. D

d.

| Correction criterion         | Points          |
|------------------------------|-----------------|
| Notion of uniquely determine | 4 points        |
| Correct                      | 3 points        |
| <i>Total points:</i>         | <i>7 points</i> |

e. A

1 pt. B

1 pt. C

1 pt. D

Bonus: 0 pt.

f. 3 pt. A

The grading of this question was changed: 2 points for a correct answer; 2 points for noticing that the constraint was not violated.

g.

| Correction criterion                          | Points          |
|-----------------------------------------------|-----------------|
| Answer correct {40,60,90,170}                 | 1 point         |
| Noticing that the constraint was not violated | 2 points        |
| <i>Total points:</i>                          | <i>4 points</i> |

h.

| Correction criterion                       | Points          |
|--------------------------------------------|-----------------|
| Answer NO                                  | 1 point         |
| At least one correct (3 or 5)              | 2 points        |
| Both correct and no wrong things mentioned | 2 points        |
| <i>Total points:</i>                       | <i>5 points</i> |

- i. A  
 1 pt. B  
 C  
 D  
 E  
 Bonus: 1 pt.

2. a. 1. 1 pt. A  
 23 pt. 2. 1 pt. E  
 3. 1 pt. D  
 4. 1 pt. B  
 5. 1 pt. C

Bonus: 2 pt.

- b. 4 pt. \*fl\*lo\*ow\*we\*er\*  
 c. 1 pt. 3  
 d. 1 pt. 6  
 e. 2 pt. 0,5

f.

| Correction criterion                                    | Points          |
|---------------------------------------------------------|-----------------|
| Application of thresholds                               | 1 point         |
| Three cases: certain match, possible, certain non-match | 1 point         |
| Combinations to produce partitionings                   | 2 points        |
| <i>Total points:</i>                                    | <i>4 points</i> |

g.

| Correction criterion                          | Points          |
|-----------------------------------------------|-----------------|
| Multiplication of probabilities               | 1 point         |
| Probabilities for certain and uncertain edges | 2 points        |
| All edges                                     | 1 point         |
| <i>Total points:</i>                          | <i>4 points</i> |

3. a. 1 pt. A  
20 pt. 1 pt. B  
C  
D  
E  
1 pt. F  
Bonus: 1 pt.

- b. 3 pt. B

c.

| Correction criterion                                                | Points          |
|---------------------------------------------------------------------|-----------------|
| What they can do (e.g., data cleaning)                              | 3 points        |
| What they cannot do (one strong reason or two weaker ones, or more) | 5 points        |
| <i>Total points:</i>                                                | <i>8 points</i> |

d.

| Correction criterion | Points          |
|----------------------|-----------------|
| Criterion            | 6 points        |
| <i>Total points:</i> | <i>6 points</i> |

## Cesuur

Toegepaste raadscore: 6,119 pt

| Behaalde punten | Cijfer |
|-----------------|--------|
| 90              | 10     |
| 89              | 9,9    |
| 88              | 9,8    |
| 87              | 9,7    |
| 86              | 9,6    |
| 85              | 9,5    |
| 84              | 9,4    |
| 83              | 9,2    |
| 82              | 9,1    |
| 81              | 9,0    |
| 80              | 8,9    |
| 79              | 8,8    |
| 78              | 8,7    |
| 77              | 8,6    |
| 76              | 8,5    |
| 75              | 8,4    |
| 74              | 8,3    |
| 73              | 8,2    |
| 72              | 8,1    |
| 71              | 8,0    |
| 70              | 7,9    |
| 69              | 7,7    |
| 68              | 7,6    |
| 67              | 7,5    |
| 66              | 7,4    |
| 65              | 7,3    |
| 64              | 7,2    |
| 63              | 7,1    |
| 62              | 7,0    |

|    |     |
|----|-----|
| 61 | 6,9 |
| 60 | 6,8 |
| 59 | 6,7 |
| 58 | 6,6 |
| 57 | 6,5 |
| 56 | 6,4 |
| 55 | 6,2 |
| 54 | 6,1 |
| 53 | 6,0 |
| 52 | 5,9 |
| 51 | 5,8 |
| 50 | 5,7 |
| 49 | 5,6 |
| 48 | 5,5 |
| 47 | 5,4 |
| 46 | 5,3 |
| 45 | 5,2 |
| 44 | 5,1 |
| 43 | 5,0 |
| 42 | 4,8 |
| 41 | 4,7 |
| 40 | 4,6 |
| 39 | 4,5 |
| 38 | 4,4 |
| 37 | 4,3 |
| 36 | 4,2 |
| 35 | 4,1 |
| 34 | 4,0 |
| 33 | 3,9 |
| 32 | 3,8 |

|    |     |
|----|-----|
| 31 | 3,7 |
| 30 | 3,6 |
| 29 | 3,5 |
| 28 | 3,3 |
| 27 | 3,2 |
| 26 | 3,1 |
| 25 | 3,0 |
| 24 | 2,9 |
| 23 | 2,8 |
| 22 | 2,7 |
| 21 | 2,6 |
| 20 | 2,5 |
| 19 | 2,4 |
| 18 | 2,3 |
| 17 | 2,2 |
| 16 | 2,1 |
| 15 | 2,0 |
| 14 | 1,8 |
| 13 | 1,7 |
| 12 | 1,6 |
| 11 | 1,5 |
| 10 | 1,4 |
| 9  | 1,3 |
| 8  | 1,2 |
| 7  | 1,1 |
| 6  | 1,0 |
| 5  | 1,0 |
| 4  | 1,0 |
| 3  | 1,0 |
| 2  | 1,0 |
| 1  | 1,0 |
| 0  | 1,0 |